# Novel Lead-Free $Cs_2NaFeBr_6$ Double Perovskite Exhibiting Multifunctional Energy-Relevant Properties: A Comprehensive First-Principles Study

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## ABSTRACT

The rapid demand for high-efficiency and environmentally benign photovoltaic absorbers has intensified the search for lead-free halide perovskites that can simultaneously deliver strong optoelectronic response and long-term stability. In this context, double perovskites offer an attractive design space due to their compositional flexibility, suppressed Pb-toxicity concerns, and improved structural robustness compared to many Pb-based counterparts. Motivated by these considerations, we investigate the cubic lead-free double perovskite $Cs_2NaFeBr_6$ using density functional theory and complementary stability and device-metric analyses. Geometric formability indicators confirm the feasibility of the cubic framework with $t = 0.9724$, $\mu = 0.425$, and $\tau = 0.425$, while thermodynamic evaluation yields a negative formation energy and a positive decomposition energy, supporting energetic stability. Phonon dispersion shows no imaginary frequencies, establishing dynamical stability. The mechanical properties confirm a mechanically stable, relatively soft halide lattice, and the elastic constants further suggest ductile behavior. The Debye temperature is estimated to be $\Theta_D \approx 254$ K, consistent with the comparatively soft bonding environment of halide frameworks. Electronic-structure calculations reveal a direct bandgap of 1.09 eV, which lies close to the optimum window for single-junction photovoltaics. Carrier-transport descriptors derived from band-edge curvature show moderate effective masses implying favorable electron transport and reasonably balanced carrier dynamics. Optical response exhibits strong dielectric screening with a static dielectric constant $\varepsilon_1(0) \approx 6.5$, supporting reduced Coulombic trapping and recombination. Finally, spectroscopic limited maximum efficiency analysis under AM 1.5G illumination predicts high photovoltaic performance, yielding a thickness-dependent power conversion efficiency (PCE) of 30.60% at 1 $\mu$m and a maximum PCE of 32.09% at 5 $\mu$m, approaching the Shockley-Queisser ceiling for this bandgap. Collectively, the stability signatures and optoelectronic metrics position $Cs_2NaFeBr_6$ as a promising lead-free absorber for high-efficiency photovoltaic devices and motivate further experimental validation.

## Introduction

The accelerating global demand for energy continues to intensify fossil-fuel consumption, amplifying greenhouse-gas emissions, air pollution, and climate-driven risks. This reality has strengthened the global push for sustainable energy conversion technologies. Although silicon photovoltaics dominate today's market, their practical efficiency ceilings and manufacturing complexity motivate the search for alternative photovoltaic (PV) absorbers that combine strong performance with scalable, low-cost processing[1]. In this context, lead halide perovskites have emerged as a transformative class of PV materials, enabled by their outstanding optical absorption, tunable bandgaps, long carrier diffusion lengths, and favorable charge-transport characteristics[2]. Consequently, perovskite solar cells have advanced rapidly, reaching a certified single-junction power conversion efficiency (PCE) of 26.7%[3]. Beyond photovoltaics, lead halide perovskites have also demonstrated broad optoelectronic utility, including lasers[4], photodetectors[5,6], transistors[7], and photocatalysis[8]. Despite these successes, two persistent issues hinder wide deployment: (i) limited long-term stability under moisture, oxygen, heat, and illumination, and (ii) the toxicity and high bioavailability of lead, which raises environmental and health concerns[9,10].

To mitigate these limitations, research has increasingly focused on stable, lead-free halide perovskite derivatives, with

cesium-based double perovskites (DPs) receiving particular attention due to their thermal robustness, structural diversity, and environmentally benign composition. Halide DPs often adopt elpasolite-derived frameworks described by the general formula $Cs_2M^{I}M^{III}X_6$, where the ionic-radius match between monovalent and trivalent cations plays a critical role in determining phase formation, octahedral ordering, and symmetry. Early crystallographic studies on chloride elpasolites $Cs_2LiM^{3+}Cl_6$ established that stable crystallization is strongly governed by the ionic radius of the trivalent metal cation[11]. Importantly, these investigations also revealed radius-driven structure selection within the elpasolite family; for example, compounds containing relatively small trivalent cations (e.g., $Cr^{3+}$) can stabilize hexagonal, two-layer elpasolite-derived structures without evidence of decomposition or polymorphic instability, underscoring the thermodynamic viability of multiple ordered DP polymorphs[11]. Motivated by this structurally accessible landscape, first-principles screening studies have explored a wide compositional space spanning alkali metals (Li, Na, K, Cs), trivalent cations (In, Sc, V, Fe, Cu, Ga), and halides (Cl, Br), aiming to identify candidates with favorable bandgaps, strong carrier transport, and high chemical stability[12–15]. Collectively, these efforts provide a consistent theoretical basis for extending lead-free DP design toward *d*-block-containing compounds, where electronic correlations and spin-dependent physics can further enrich functionality.

Within solar-energy conversion, Cs-based lead-free DPs have demonstrated promising performance and stability across photovoltaics and photocatalysis. The prototypical $Cs_2AgBiBr_6$ system, for instance, is widely recognized for its robust environmental stability, high crystallinity, and long carrier lifetimes. Its optoelectronic response can be tuned through chemical modifications, including hydrogenation, yielding reported bandgap reductions down to $\sim$1.64 eV and PV efficiencies of several percent in device-oriented studies[16–19]. Beyond PV operation, DP-based photocatalysts have shown encouraging activity for hydrogen evolution and $CO_2$ reduction. For example, $Cs_2AgBiBr_6$-based composites can achieve high selectivity toward CO formation under visible irradiation, while $Cs_2NaBiBr_6$ has been reported with low kinetic barriers and pronounced facet-dependent catalytic behavior[20,21]. Related DP compositions such as $Cs_2NaBiCl_6$ combine favorable optical and thermoelectric attributes with robust stability, and Cu-doped $Cs_2AgSbCl_6$ microcrystals have been experimentally validated as efficient photocatalysts for $CO_2$ reduction[22–24]. These representative results collectively establish Cs-based double perovskites as multifunctional, stable platforms for light harvesting and solar-fuel-related processes.

A particularly compelling direction is the incorporation of transition-metal cations into halide DPs, enabling coupled magnetic, electronic, and optoelectronic behaviors within a chemically stable lattice. Transition-metal-containing families such as $Cs_2AgT^{3+}Cl_6$ (T = Fe, Cr)[25,26], $Cs_2NaT^{3+}Cl_6$ (T = Fe, V, Mn, Ni)[27,28], and $Cs_2KT^{3+}Cl_6$ (T = Mn, Co, Ni)[29] exhibit diverse magnetic orderings and spin-mediated electronic responses. For example, hexagonal $Cs_2AgCrCl_6$ has been synthesized in a paramagnetic phase and exhibits attractive optoelectronic characteristics[26], while cubic $Cs_2AgFeCl_6$ has demonstrated encouraging photovoltaic relevance[25]. More recently, $Cs_2NaFeCl_6$ single crystals have been reported to show high thermal stability and reversible thermochromism from light yellow to black over 10–423 K[30]. These findings highlight $Fe^{3+}$-containing elpasolite-type DPs as functional materials where lattice stability, electronic structure, and spin physics can converge. In parallel, first-principles studies on bromide elpasolites such as $Cs_2LiGaBr_6$ and $Cs_2NaGaBr_6$ predict visible-light bandgaps near $\sim$1.8 eV, strong electron–phonon interactions, high optical absorption, and thermoelectric figures of merit exceeding unity, emphasizing that bromide-based $Cs_2M^{I}M^{III}Br_6$ chemistries can be highly promising for integrated energy-conversion functionalities[31].

Motivated by this landscape, the present work focuses on the bromide analogue $Cs_2NaFeBr_6$, a Cs-based transition-metal double perovskite that combines an earth-abundant $Fe^{3+}$ center with a chemically stable elpasolite-type framework. Relative to the chloride counterpart $Cs_2NaFeCl_6$, which already exhibits robust thermal stability and thermochromic behavior[30], substitution of $Cl^-$ by the heavier and more polarizable $Br^-$ anion is expected to modify metal-halide covalency, tune the band-edge chemistry, and potentially enhance visible-light interaction features that are directly relevant for photocatalytic and optoelectronic applications. At the same time, transition-metal *d*-states and correlation effects can play a decisive role in determining band dispersion, carrier masses, and spin-dependent electronic structure, necessitating a careful first-principles treatment consistent with prior DP screening and design studies[12–15]. In this study, we therefore perform a comprehensive DFT-based investigation of $Cs_2NaFeBr_6$, assessing its structural stability, mechanical robustness, electronic band characteristics, and carrier effective masses. Through this analysis, we aim to clarify the structure–property relationships of $Cs_2NaFeBr_6$ and position it as a stable and environmentally responsible candidate for future energy-related optoelectronic and photocatalytic technologies.

## Results and Discussion

### Crystal Structure

The double perovskite $Cs_2NaFeBr_6$ adopts a high-symmetry cubic structure belonging to the $Fm\bar{3}m$ space group. As an $A_2BB'X_6$ type framework, it consists of a corner-sharing octahedral network where the large $Cs^+$ ions occupy the 8*c* Wyckoff sites. The *B* and *B′* sites are alternately populated by $Na^+$ (4*a*) and $Fe^{3+}$ (4*b*) cations, respectively, forming a perfectly ordered rock-salt configuration. Within this lattice, the $Br^-$ ions reside on the 24*e* sites, defining the coordination environment for

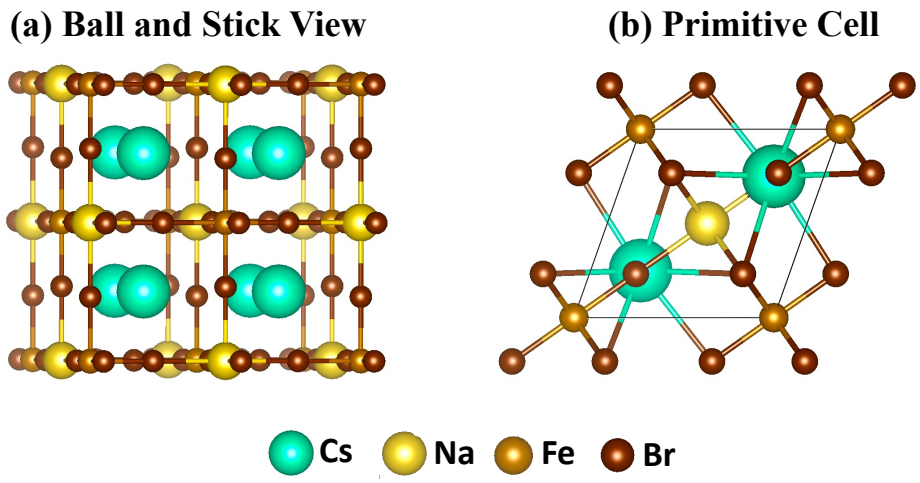


**Figure 1.** Crystal structure of $Cs_2NaFeBr_6$. (a) Ball-and-stick representation of the conventional unit cell. (b) Corresponding primitive cell.

the central cations as depicted in Fig 1. Based on the calculations, the material exhibits an equilibrium lattice constant of 10.837 Å. The internal geometry is defined by distinct bond lengths: the Fe–Br bond measures 2.235 Å, while the Na–Br bond is significantly longer at 2.883 Å. This disparity in cation size is further reflected in the polyhedral dimensions, with the edge length of the Na polyhedron measuring 4.078 Å compared to the 3.586 Å edge length of the Fe polyhedron.

Despite the difference in cation radii, the framework remains remarkably rigid; the Fe–Br–Na bond angle is exactly 180°. This linear arrangement confirms that the adjacent $[FeBr_6]$ and $[NaBr_6]$ octahedra are aligned without any tilting or rotation. Such structural stability preserves the ideal cubic symmetry and ensures a continuous, undistorted Br-bridged connectivity throughout the lattice.

### Structural and Thermodynamic Stability

The stability of $Cs_2NaFeBr_6$ was evaluated from both geometric (structural) and energetic (thermodynamic) perspectives to confirm that the compound can sustain an undistorted cubic double-perovskite framework. Halide double perovskites with the general stoichiometry $A_2BB'X_6$ ideally crystallize in the cubic $Fm\bar{3}m$ structure; however, departures from the ideal arrangement may occur due to ionic size mismatch, octahedral tilting, lattice strain, or Jahn–Teller-related distortions. Therefore, prior to analyzing functional properties, it is essential to validate that $Cs_2NaFeBr_6$ satisfies both geometric formability criteria and energetic stability requirements.

**Structural stability.** To examine whether the cubic lattice of $Cs_2NaFeBr_6$ is geometrically favorable and resistant to structural deformation, three widely used stability descriptors were employed: the Goldschmidt tolerance factor ($t$)[32], the octahedral factor ($\mu$)[33], and the recently introduced tolerance descriptor ($\tau$)[34]. The tolerance factor $t$ captures the size compatibility between the A-site cation and the $BX_6$ octahedral framework, the octahedral factor $\mu$ assesses the ability of the B/B′ cations to be accommodated inside the $X_6$ octahedron, and the $\tau$ descriptor provides a generalized geometric criterion that better captures deviations from the ideal cubic geometry through nonlinear size-ratio dependence. These indices are defined as:

$$t = \frac{r_A + r_X}{\sqrt{2}\left(\frac{r_B + r_{B'} + 3r_X}{2}\right)} \tag{1}$$

$$\mu = \frac{r_B + r_{B'}}{2r_X} \tag{2}$$

$$\tau = \frac{r_X}{r_B} - n_A\left(n_A - \frac{r_A/r_B}{\ln(r_A/r_B)}\right) \tag{3}$$

Here, $r_A$, $r_B$, $r_{B'}$, and $r_X$ represent the ionic radii of the ions occupying the A, B, B′, and X sites, respectively. For $Cs_2NaFeBr_6$, these correspond to $Cs^+$ (A site), $Na^+$ (B site), $Fe^{3+}$ (B′ site), and $Br^-$ (X site). The ionic radii were taken from

Shannon's effective ionic radii database[35], selected according to the oxidation state and coordination number consistent with the DFT-optimized cubic environment: $Cs^+$ in 12-fold coordination and $Na^+$, $Fe^{3+}$, and $Br^-$ in sixfold coordination.

For a stable three-dimensional perovskite framework, $t$ typically lies between 0.80 and 1.00 (with slightly higher values up to $\sim$1.10 sometimes reported for distorted or layered variants). Values below this window are commonly associated with octahedral tilting and symmetry lowering, whereas values significantly above unity often favor non-perovskite structures. The octahedral factor should satisfy $\mu \geq 0.414$ ($\approx \sqrt{2}-1$) to ensure geometric stability of the $BX_6$ octahedra. Although a universally fixed stability window for $\tau$ is not yet established, lower $\tau$ values generally correlate with stable three-dimensional perovskite frameworks. In $Cs_2NaFeBr_6$, the computed values ($t = 0.9724$, $\mu = 0.425$, and $\tau = 0.425$) fall within or very close to the ranges typically reported for stable cubic double perovskites, indicating that the compound is geometrically favorable for forming a robust cubic $Fm\bar{3}m$ lattice without pronounced octahedral tilting or distortion.

To further rationalize these results, the geometric descriptors of $Cs_2NaFeBr_6$ were compared with compositionally similar halide double perovskites reported to crystallize in the same cubic ($Fm\bar{3}m$) structure. Representative examples include $Cs_2NaFeCl_6$, $Cs_2AgBiBr_6$, $Cs_2NaBiCl_6$, and $Cs_2AgInCl_6$, which are known to form stable cubic frameworks within comparable tolerance and octahedral factor ranges. Xing *et al.*[36] reported a tolerance factor close to 1.00 for $Cs_2NaFeCl_6$. McClure *et al.*[37] and Slavney *et al.*[38] showed that $Cs_2AgBiBr_6$ and $Cs_2NaBiCl_6$ remain cubic for $t$ values between 0.94-0.98 and $\mu$ values around 0.43-0.45. Likewise, Volonakis *et al.*[39] confirmed a stable cubic $Fm\bar{3}m$ phase for $Cs_2AgInCl_6$, exhibiting geometric parameters within the same range as those obtained for $Cs_2NaFeBr_6$. The close agreement with these established cubic double perovskites further supports the structural viability of $Cs_2NaFeBr_6$.

**Thermodynamic stability.** In addition to geometric considerations, thermodynamic stability was assessed through the formation energy ($E_F$) per atom at 0 K and the decomposition energy ($E_D$). The formation energy quantifies energetic favorability relative to the elemental reference phases and was computed as:

$$E_F = \frac{E_{Cs_2NaFeBr_6} - 2E_{Cs} - E_{Na} - E_{Fe} - 6E_{Br}}{10} \tag{4}$$

where $E_{Cs_2NaFeBr_6}$ is the total DFT energy of $Cs_2NaFeBr_6$, and $E_{Cs}$, $E_{Na}$, $E_{Fe}$, and $E_{Br}$ represent the total energies of elemental Cs, Na, Fe, and Br in their standard states. The denominator (10) corresponds to the total number of atoms in one formula unit (2 Cs + 1 Na + 1 Fe + 6 Br). A negative $E_F$ indicates that compound formation is energetically favorable with respect to elemental segregation. To examine stability against decomposition into likely competing bromide phases, the following reaction pathway was considered:

$$Cs_2NaFeBr_6 \rightarrow 2CsBr + NaBr + FeBr_3 \tag{5}$$

Accordingly, the decomposition energy was calculated as:

$$E_D = 2E_{CsBr} + E_{NaBr} + E_{FeBr_3} - E_{Cs_2NaFeBr_6} \tag{6}$$

Here, $E_{CsBr}$, $E_{NaBr}$, $E_{FeBr_3}$, and $E_{Cs_2NaFeBr_6}$ denote the DFT total energies of CsBr, NaBr, $FeBr_3$, and $Cs_2NaFeBr_6$, respectively. In general, a negative formation energy and a positive decomposition energy indicate thermodynamic stability against elemental decomposition and phase separation into competing bromide phases[40,41]. For $Cs_2NaFeBr_6$, the calculated values ($E_F = -1.27$ eV/atom and $E_D = 0.77$ eV/atom) therefore confirm that the compound is energetically stable. The Shannon ionic radii and the derived geometric ($t$, $\mu$, and $\tau$) and thermodynamic ($E_F$ and $E_D$) stability metrics are summarized in Table 1.

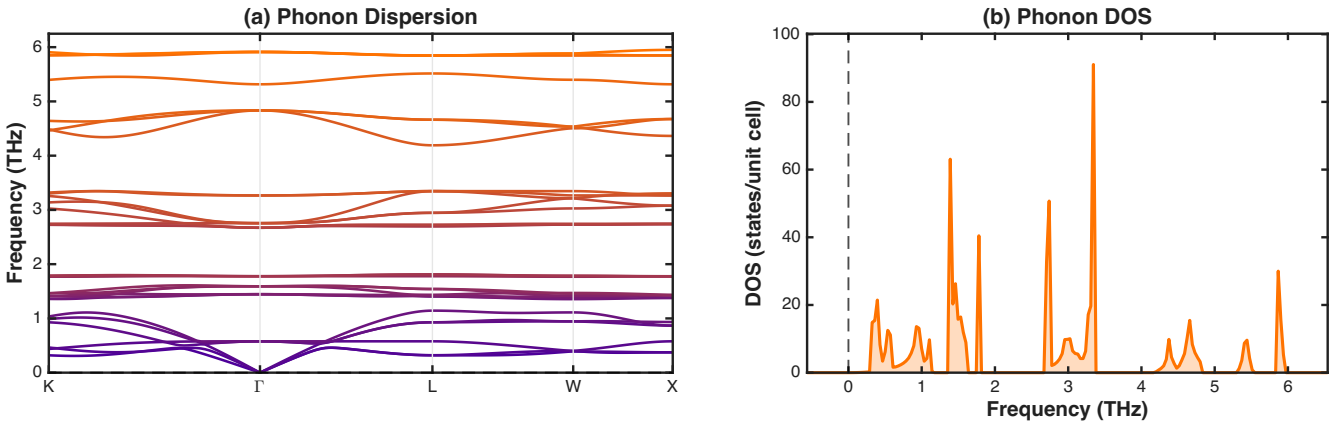


**Figure 2.** Phonon band structure and total phonon density of states (TDOS) of $Cs_2NaFeBr_6$. (a) Phonon dispersion relations along the selected high-symmetry directions in the Brillouin zone. (b) Corresponding total phonon DOS, highlighting the contributions of different vibrational modes across the frequency range.

**Table 1.** Shannon effective ionic radii (selected according to oxidation state and coordination environment) and calculated geometric and thermodynamic stability indices for $Cs_2NaFeBr_6$.

| Quantity | Parameter | Ionic State | Coord. No. | Value |
|---|---|---|---|---|
| Ionic radii (pm) | Cs ($r_{Cs} = r_A$) | +1 | 12 | 188 |
| | Na ($r_{Na} = r_B$) | +1 | 6 | 102 |
| | Fe ($r_{Fe} = r_{B'}$) | +3 | 6 | 64.5 |
| | Br ($r_{Br} = r_X$) | −1 | 6 | 196 |
| Stability indices | Goldschmidt tolerance factor ($t$) | – | – | 0.9724 |
| | Octahedral factor ($\mu$) | – | – | 0.425 |
| | Tau tolerance descriptor ($\tau$) | – | – | 1.865 |
| | Formation energy $E_F$ (eV/atom) | – | – | −1.27 |
| | Decomposition energy $E_D$ (eV) | – | – | 0.77 |

## Dynamic Stability

The lattice dynamical stability of the double perovskite $Cs_2NaFeBr_6$ was investigated by computing the phonon dispersion along high-symmetry directions of the Brillouin zone, as shown in Fig. 2, using density functional theory within the VASP–Phonopy framework. A primitive-cell description was adopted to accurately resolve the vibrational spectrum and associated phonon interactions. $Cs_2NaFeBr_6$ contains 10 atoms in the primitive unit cell, which gives rise to 30 phonon branches in total: three acoustic modes (one longitudinal acoustic, LA, and two transverse acoustic, TA) and 27 optical modes. The zone-center vibrational representation at the Γ point can be expressed as:

$$\Gamma = 10E_u + 5A_{2u} + 9T_g + 3T_u + 2E_g + A_{1g}. \tag{7}$$

Here, $A$, $E$, and $T$ denote non-degenerate, doubly degenerate, and triply degenerate irreducible representations, respectively. The acoustic contribution at Γ consists of two transverse acoustic branches ($2E_u$) and one longitudinal acoustic branch ($A_{2u}$), which converge to zero frequency, as required by translational invariance. Importantly, no imaginary phonon frequencies are observed throughout the Brillouin zone, confirming the dynamical stability of $Cs_2NaFeBr_6$ against spontaneous lattice instabilities and structural distortions[42,43]. These acoustic branches are optically inactive because they correspond to rigid translations of the lattice and therefore do not generate changes in either the macroscopic dipole moment or the polarizability tensor.

The longitudinal optical phonon branches are mainly distributed within three frequency windows: ∼0–2 THz, 2.8–4.8 THz, and 5–6 THz. In the low-frequency region, the vibrational dynamics are dominated by Cs-related motions, which can be primarily associated with wagging-type vibrations involving Cs and Br atoms. In the intermediate-frequency range, the modes arise predominantly from coupled vibrations of Br atoms with the two metal cations (Na and Fe). The highest-frequency region is mainly characterized by Fe–Br stretching vibrations, leading to a comparatively lower density of phonon states. Overall, the

absence of imaginary modes and the well-separated frequency regions demonstrate that $Cs_2NaFeBr_6$ is dynamically stable and can sustain lattice vibrations without triggering spontaneous structural distortions.

## Mechanical Properties

The mechanical properties and failure modes of a material determine how it responds to externally applied loads and therefore represent key parameters for assessing its suitability in practical applications. Investigation of these properties provides valuable information about bonding characteristics, elastic stability, and mechanical robustness[44]. For the double perovskite $Cs_2NaFeBr_6$, the mechanical response was analyzed using the strain–energy method, where a series of small deformations is imposed on the equilibrium crystal lattice and the total energies of the resulting distorted configurations are calculated. The second-order elastic constants are then obtained from the strain dependence of the total energy and serve as primary indicators of mechanical performance.

The elastic response in the linear regime is governed by Hooke's law and is conveniently treated within the Voigt–Reuss–Hill (VRH) approximation, which provides a systematic framework to estimate polycrystalline elastic properties. In cubic crystals, only three independent elastic stiffness constants, $C_{11}$, $C_{12}$, and $C_{44}$, are sufficient to describe the full elastic behavior and mechanical stability[45]. The calculated elastic constants are all positive, indicating that $Cs_2NaFeBr_6$ satisfies the generalized Born–Huang mechanical stability criteria[46]:

$$C_{11} - C_{12} > 0, \quad C_{11} + 2C_{12} > 0, \quad C_{44} > 0, \quad C_{11} > 0, \quad C_{12} < B < C_{11}.$$

Within the VRH formalism, the shear modulus in the vVoigt limit and the Reuss limit, as well as the bulk modulus for cubic systems, are expressed as[47]

$$G_V = \frac{C_{11} - C_{12} + 3C_{44}}{5}, \qquad \frac{5}{G_R} = \frac{4}{C_{11} - C_{12}} + \frac{3}{C_{44}}, \qquad B = B_V = B_R = \frac{C_{11} + 2C_{12}}{3}.$$

The Hill-averaged shear modulus is obtained from $G = (G_V + G_R)/2$, and the Young's modulus and Poisson's ratio are then determined using[48]

$$E = \frac{9BG}{3B + G}, \qquad \nu = \frac{3B - 2G}{2(3B + G)}.$$

To quantify elastic anisotropy, the Zener anisotropy factor is defined as $A = 2C_{44}/(C_{11} - C_{12})$[48]. The computed VRH-averaged elastic moduli and associated mechanical descriptors are summarized in Table 2[42,46].

**Table 2.** Summary of elastic/mechanical properties of $Cs_2NaFeBr_6$ and comparison with related Cs-based double perovskites from the literature, including Debye temperature $\Theta_D$ and estimated melting point $T_m$ where available.

| **Compound** | $C_{11}$ **(GPa)** | $C_{44}$ **(GPa)** | $C_{12}$ **(GPa)** | $B$ **(GPa)** | $G$ **(GPa)** | $E$ **(GPa)** | $\nu$ | $B/G$ | $\Theta_D$ / $T_m$ **(K / K)** | **Ref.** |
|---|---|---|---|---|---|---|---|---|---|---|
| $Cs_2NaFeBr_6$ (Voigt) | 36.642 | 8.532 | 10.127 | 18.97 | 10.42 | 26.43 | 0.270 | 1.82 | 254 / 613 | This work |
| $Cs_2NaFeBr_6$ (Reuss) | 36.642 | 8.532 | 10.127 | 18.965 | 9.951 | 25.409 | 0.277 | 1.906 | 254 / 613 | This work |
| $Cs_2NaFeBr_6$ (Hill) | 36.642 | 8.532 | 10.127 | 18.965 | 10.181 | 25.919 | 0.272 | 1.862 | 254 / 613 | This work |
| $Cs_2NaBiBr_6$ | – | – | 11.50 | 18.20 | 10.10 | 25.10 | 0.27 | 1.84 | – / 621 | 49 |
| $Cs_2NaBiI_6$ | – | – | 9.90 | 16.10 | 9.80 | 24.60 | 0.24 | 1.63 | – / 611.5 | 49 |
| $Cs_2NaTiBr_6$ | – | – | 7.726 | 14.607 | 7.493 | 19.197 | 0.275 | 1.94 | 130.80 / 598.66 | 50 |
| $Cs_2NaGaBr_6$ | – | – | – | 22.80 | 13.64 | 34.11 | 0.25 | 1.65 | – / – | 51 |
| $Cs_2NaCrCl_6$ | – | – | 17.23 | 27.15 | 16.29 | 40.725 | 0.74 | 1.66 | – / 654.84 | 52,53 |
| $Cs_2NaTiCl_6$ | – | – | 10.218 | – | – | – | – | – | 172.937 / 614.20 | 50 |
| $Cs_2NaTiF_6$ | – | – | 23.440 | – | – | – | – | – | 237.45 / 691.55 | 50,53 |

Young's modulus $E$ is one of the most widely used mechanical descriptors of stiffness, as it measures how strongly a material resists elastic stretching or compression along an applied tensile direction. In the linear elastic region, it is defined as the ratio of tensile stress to tensile strain[43], meaning that a material with a larger $E$ requires a higher stress to produce the same amount of elastic deformation. Therefore, higher values of $E$ indicate a stiffer lattice and greater resistance to reversible (elastic) distortion. Using the VRH-derived elastic moduli and the standard relation $E = 9BG/(3B + G)$, the Young's modulus of $Cs_2NaFeBr_6$ is obtained as $E = 26.43$ GPa, indicating a moderate stiffness typical of relatively soft halide-perovskite frameworks.

In addition to $E$, the bulk modulus $B$ and shear modulus $G$ provide complementary insight into how the crystal responds to different types of mechanical loading. The bulk modulus $B$ quantifies resistance to uniform volume compression under

hydrostatic pressure and is closely related to the equilibrium bonding strength and compressibility of the crystal lattice: a larger $B$ generally reflects stronger average interatomic interactions and lower compressibility. The shear modulus $G$, on the other hand, describes the resistance to shape change (distortion) at nearly constant volume under shear stress, and it is therefore a direct indicator of rigidity against lattice shearing and angular distortions. In general, higher values of $B$ and $G$ imply stronger resistance to compression and shear deformation, respectively, and suggest improved mechanical robustness. For $Cs_2NaFeBr_6$, the evaluated moduli are $B = 18.97$ GPa and $G = 10.18$ GPa, demonstrating a comparatively compressible yet mechanically stable lattice where resistance to shear deformation remains moderate.

A broader comparison of elastic response across Cs-based double perovskites is summarized in Table 2, which reveals notable composition-dependent variations in stiffness and rigidity. Among the compared compounds, $Cs_2NaCrCl_6$ exhibits the highest stiffness with a reported Young's modulus of $E = 40.73$ GPa[52], indicating a substantially more rigid lattice, while $Cs_2NaGaBr_6$ also shows enhanced stiffness with $E = 34.11$ GPa[51]. In contrast, $Cs_2NaBiBr_6$ and $Cs_2NaBiI_6$ display moderate stiffness, with Young's moduli lying within the range of 24.6–26.43 GPa[49], reflecting comparatively softer lattices. Within this context, $Cs_2NaFeBr_6$ shows a higher stiffness than $Cs_2NaBiI_6$[49] and $Cs_2NaTiBr_6$[50], while remaining comparable to $Cs_2NaBiBr_6$[49]. This overall trend suggests that $Cs_2NaFeBr_6$ maintains a balanced elastic response among the examined compositions and offers a moderate resistance to deformation while preserving mechanical stability.

To assess whether a material exhibits ductile or brittle behavior, Poisson's ratio $\nu$ is commonly employed as a practical indicator, with a widely used critical threshold at $\nu = 0.26$[45]. Materials with $\nu > 0.26$ are generally classified as ductile, whereas those with $\nu < 0.26$ are considered brittle[45]. In parallel, the balance between brittleness and toughness can be evaluated using Pugh's ratio, defined as the ratio of bulk modulus to shear modulus, $B/G$. This parameter also serves as a measure of mechanical flexibility, with a critical value of $B/G = 1.75$ separating brittle ($B/G < 1.75$) from ductile ($B/G > 1.75$) responses[45]. For $Cs_2NaFeBr_6$, the calculated Poisson's ratio exceeds the critical limit, indicating a ductile response. Consistently, the VRH-averaged values of $B/G = 1.862$ and $\nu = 0.272$ further support ductility at $P = 0$ GPa and $T = 0$ K. This classification is also reinforced by the Frantsevich criterion, which is often expressed using the inverse ratio $G/B$, where materials are considered brittle when the corresponding value falls below 0.26 and ductile otherwise[54]. For $Cs_2NaFeBr_6$, $G/B = 0.57$, which again indicates ductile behavior under ambient conditions.

Elemental substitution provides an effective route for tuning this ductility–rigidity balance in Cs-based double perovskites. Substituting Fe with Bi leads to a modest reduction in both bulk and shear moduli while maintaining comparable ductility, suggesting that Bi incorporation causes only minor perturbation to the elastic framework[49]. In contrast, $Cs_2NaBiI_6$ exhibits a further softening tendency, and the accompanying decrease in $B/G$ reflects a comparatively softer lattice relative to $Cs_2NaFeBr_6$[49]. Replacement of Fe with Ti in $Cs_2NaTiBr_6$ results in an additional reduction in both $B$ and $G$, alongside a slight increase in ductility, emphasizing the influence of lighter transition-metal substitution on the elastic response[50]. On the other hand, $Cs_2NaCrCl_6$ exhibits higher bulk, shear, and elastic moduli[52], consistent with stronger bonding and a stiffer lattice compared with $Cs_2NaFeBr_6$ and $Cs_2NaGaBr_6$[51]. These trends highlight that $Cs_2NaFeBr_6$ maintains a balanced combination of stiffness and ductility among the studied compositions, while compositional engineering can systematically shift the mechanical response toward enhanced flexibility or increased rigidity. The ductile nature of $Cs_2NaFeBr_6$ is further supported by the Cauchy pressure, defined as $C_p = C_{12} - C_{44}$[45]. In this criterion, a positive $C_p$ generally indicates elastic stability and a predominance of ionic interactions, whereas a negative value suggests more covalent character and a tendency toward brittleness[45]. For $Cs_2NaFeBr_6$, the calculated Cauchy pressure is positive ($C_p = 1.6$ GPa), confirming elastic stability and indicating that ionic bonding contributions dominate the mechanical response.

For atomic positions that are not fully constrained by symmetry, the Kleinman parameter $\zeta$ is an important descriptor of internal strain relaxation because it characterizes the relative displacement between the cation and anion sublattices during volume-conserving deformations[55]. It is defined as[56]

$$\zeta = \frac{C_{11} + 8C_{12}}{7C_{11} + 2C_{12}}.$$

The calculated value $\zeta = 0.50$ indicates a balanced internal relaxation mechanism, implying comparable tendencies toward bond-angle bending and bond-length stretching in response to shear-type strains.

The Debye temperature $\theta_D$ provides a direct connection between elastic and thermodynamic behavior and is closely associated with physical properties such as specific heat capacity, melting temperature, thermal expansion, and thermal conductivity[57]. In this work, $\theta_D$ was evaluated using

$$\theta_D = \frac{\hbar}{k_B}\left(\frac{3nN_A\rho}{4\pi M}\right)^{1/3} v_m,$$

where $\hbar$ is Planck's constant, $k_B$ is Boltzmann's constant, $N_A$ is Avogadro's number, $\rho$ is the density, $n$ is the number of atoms per formula unit, and $M$ is the molecular weight. The average sound velocity $v_m$ is obtained from the longitudinal ($v_l$) and

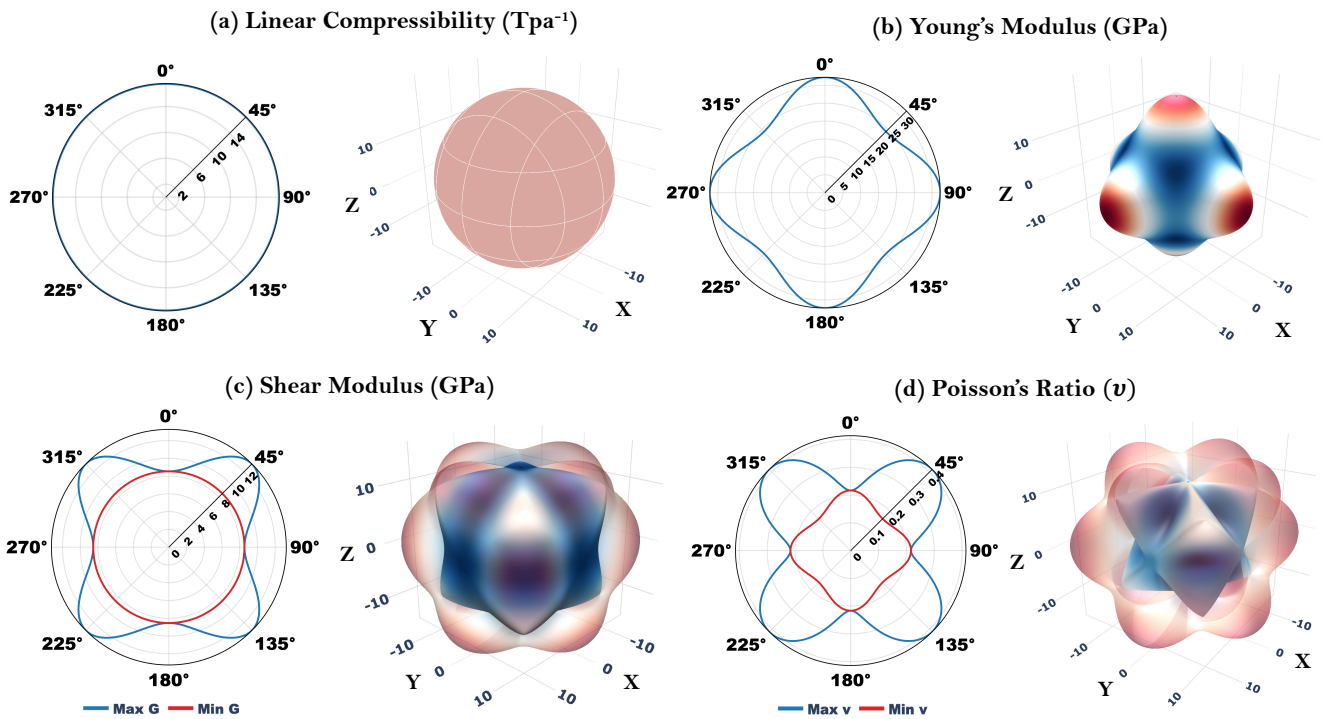


**Figure 3.** Spatial dependence of Young's modulus, Poisson's ratio, shear modulus, and linear compressibility of $Cs_2NaFeBr_6$ shown in three dimensions and as a two-dimensional polar projection on the (001) plane.

transverse ($v_t$) sound velocities according to[58]

$$v_m = \left[\frac{1}{3}\left(\frac{2}{v_l^3} + \frac{1}{v_t^3}\right)\right]^{-1/3},$$

with

$$v_t = \sqrt{\frac{G}{\rho}}, \qquad v_l = \sqrt{\frac{B + \frac{4}{3}G}{\rho}}.$$

Because phonon vibrational modes dominate heat transport in crystals, the Debye temperature is directly related to phonon–phonon scattering processes and therefore provides insight into lattice thermal conductivity. In addition, $\theta_D$ is frequently discussed in connection with the dynamic behavior of dislocations through the density of thermal phonons, and variations in $\theta_D$ can influence dislocation–electron interactions and quantum effects in solids under certain conditions[59–61]. The obtained sound velocities are $v_t = 1539.188$ m s$^{-1}$ and $v_l = 2751.302$ m s$^{-1}$, yielding an average wave velocity of $v_m = 1713.325$ m s$^{-1}$ and a Debye temperature of $\theta_D = 254$ K. As summarized in Table 2, the Debye temperatures of the investigated Cs-based double perovskites are all well below 400 K, which generally indicates relatively low phonon velocities and reduced lattice thermal conductivity[50]. The calculated Debye temperature is also lower than that of the halide perovskite $CsPbI_3$[62], reflecting a softer lattice and comparatively weaker average interatomic bonding relative to the stronger Pb–I bonding network in $CsPbI_3$[62].

The melting temperature, $T_m$, represents the temperature at which a solid phase transforms into a liquid phase, and it is often used as a practical indicator of thermal robustness. In the present work, $T_m$ was estimated from the elastic constant $C_{12}$ using the empirical relation[43]

$$T_m(\mathrm{K}) = 553 + (5.911 \times C_{12}),$$

which yields $T_m \approx 612.86$ K ($\sim 613$ K). For comparison, the reported melting temperature of $CsPbI_3$ is around 753 K[63], indicating a higher thermal resistance for $CsPbI_3$ relative to the present double perovskite. This difference can be associated with the distinct bonding environment and lattice cohesion, where the more ionic constituents and comparatively weaker average bonding in $Cs_2NaFeBr_6$ lead to a lower resistance against thermally driven disordering.

A comparative view of melting behavior across Cs-based double perovskites is summarized in Table 2. The melting temperature of $Cs_2NaFeBr_6$ is higher than those of $Cs_2NaTiBr_6$ and $Cs_2NaBiI_6$, while remaining closely comparable to $Cs_2NaTiCl_6$. In contrast, compounds containing lighter halides such as $Cs_2NaCrCl_6$ and $Cs_2NaTiF_6$ exhibit higher melting temperatures, which is commonly attributed to increased lattice energy arising from shorter metal–halide bond lengths and enhanced bond ionicity[53]. The comparatively favorable melting point is also consistent with a stable lattice configuration and strong Na–Br interactions[49], which can suppress thermal disorder and support improved thermal stability relative to more weakly bound analogs such as $Cs_2NaBiI_6$.

To further evaluate mechanical robustness, the eigenvalues of the stiffness matrix were examined, since these characteristic values describe the resistance of the lattice to different deformation modes. The stiffness-matrix eigenvalues are listed in Table 3. The three identical eigenvalues $\lambda_1$, $\lambda_2$, and $\lambda_3$ reflect a uniform elastic response to compression/extension-type deformations along different crystallographic directions, indicating a consistent baseline stiffness under axial loading. The eigenvalues $\lambda_4$ and $\lambda_5$ characterize resistance to shear-type deformation modes, where larger values correspond to stronger opposition to shearing forces, while the largest eigenvalue $\lambda_6$ represents resistance against more complex multidirectional distortions. Collectively, the distribution of these eigenvalues indicates stable resistance across compression, shear, and combined loading modes, and higher eigenvalues imply enhanced stiffness and improved resistance to deformation under the corresponding strain patterns[64].

**Table 3.** Stiffness Matrix Eigenvalues of $Cs_2NaFeBr_6$

| $\lambda_1$ | $\lambda_2$ | $\lambda_3$ | $\lambda_4$ | $\lambda_5$ | $\lambda_6$ |
|---|---|---|---|---|---|
| 8.532 GPa | 8.532 GPa | 8.532 GPa | 26.515 GPa | 26.515 GPa | 56.896 GPa |

Elastic anisotropy was quantified using the Zener anisotropy factor $A$, where $A = 1$ corresponds to an elastically isotropic solid and deviations from unity indicate anisotropy[45]. The calculated value $A = 0.643$ confirms that the elastic response is anisotropic, which is further evidenced by the strong directional dependence of the mechanical properties. Direction-resolved visualizations were generated using ELATE software[65], which provides three-dimensional representations together with two-dimensional polar projections on the (001) plane for Young's modulus, Poisson's ratio, shear modulus, and linear compressibility. All of these distributions are presented in Figure 3, where the anisotropic nature of the elastic response can be clearly observed. In particular, the Young's modulus exhibits its maximum values along the principal crystallographic axes [001], [010], and [100], while the minimum value appears along the [111] direction, confirming a pronounced directional dependence of stiffness.

## Electronic Properties

### *Band Structure*

Electronic band dispersion is central to identifying whether a material behaves as an insulator, semiconductor, semimetal, or metal, and it critically influences carrier generation, transport, and recombination in device operation. Figure 4 presents the spin-resolved band structures of $Cs_2NaFeBr_6$ calculated using the GGA-PBE and GGA-PBE+$U$ approaches along the high-symmetry directions of the Brillouin zone. The evident spin dependence indicates that exchange splitting associated with the magnetic Fe sublattice lifts the spin degeneracy and produces distinct electronic manifolds for the two spin channels. Within GGA-PBE, $Cs_2NaFeBr_6$ exhibits a pronounced spin-dependent electronic structure. The spin-up channel shows a wide band gap of 3.6275 eV, whereas the spin-down channel displays a much smaller gap of 0.5532 eV. This asymmetry can be understood in terms of the different contributions of Fe 3$d$ states hybridized with Br 4$p$ orbitals near the band edges (see Figure 5): the crystal-field environment of the $FeBr_6$ octahedra splits the 3$d$ levels, while exchange interactions further separate majority and minority spin states. Consequently, the valence and conduction edges become strongly spin selective, implying that electrons and holes can be preferentially generated in different spin channels. Because the band edges originate from different spin channels, the resulting fundamental gap is reduced to 0.4163 eV, highlighting a spin-dependent band-edge alignment that can facilitate highly spin-polarized carrier populations under electrical bias or optical excitation.

Upon including on-site correlation effects through the Hubbard $U$ correction on Fe 3$d$ states, the semiconducting character is reinforced and the gap is widened. The spin-down channel yields an indirect gap of 1.0904 eV (with a direct gap of 1.1172 eV at $\Gamma$), while the spin-up channel remains wide-gap with 3.7647 eV. Physically, the $U$ term penalizes partial occupancy and reduces the self-interaction error inherent to semilocal functionals, pushing occupied Fe 3$d$ states downward and unoccupied 3$d$ states upward in energy. This increased separation improves the description of localized $d$ electrons and strengthens the charge-transfer character of the gap, while also sharpening the exchange-split band edges. The indirect nature of the smaller gap further suggests momentum-sensitive carrier transitions, which can influence radiative recombination rates and carrier lifetimes. Overall, the persistence of a wide-gap channel alongside a narrower-gap channel indicates robust spin polarization at the band edges, a desirable feature for spin-selective transport where one spin channel can dominate conduction. This strong spin dependence in the electronic structure is therefore attractive for spintronic and spin-optoelectronic applications, enabling

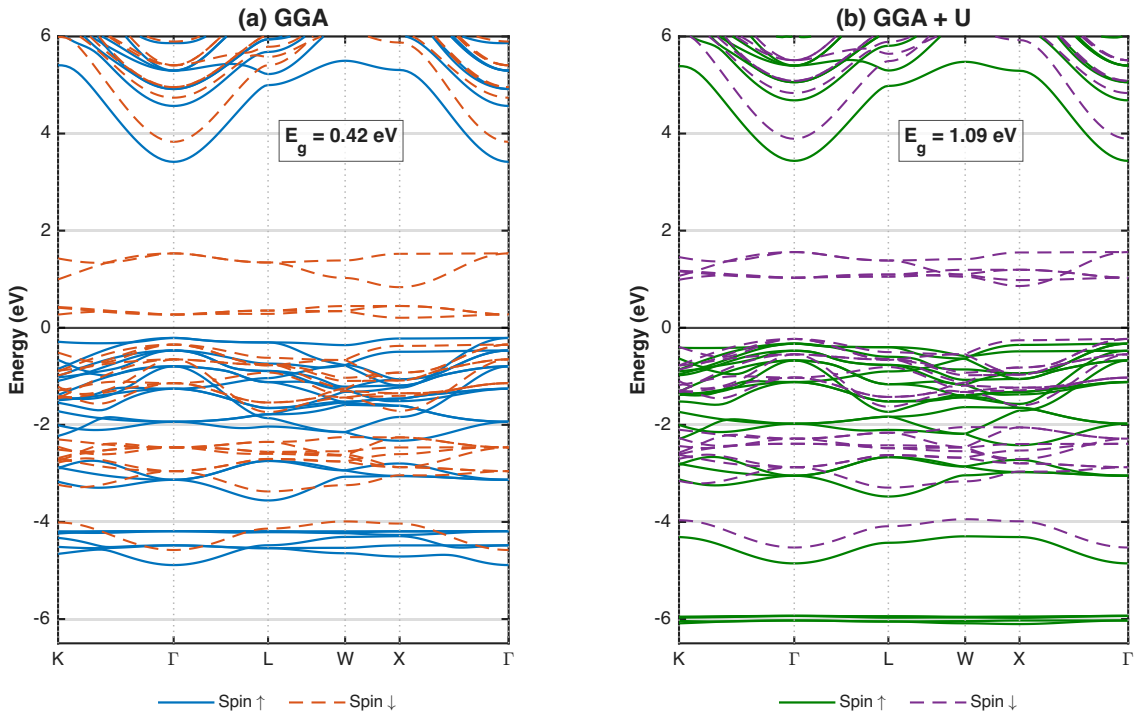


**Figure 4.** Electronic band structure of $Cs_2NaFeBr_6$: (a) GGA and (b) GGA+$U$. The Fermi level is shifted to 0 eV.

spin-polarized carrier generation and transport and supporting the potential of $Cs_2NaFeBr_6$ for device concepts that couple magnetic order with electronic and optical functionality[66–68].

### *Density of States*

To elucidate the bonding character and the orbital contributions governing the band edges of $Cs_2NaFeBr_6$, we analyzed the spin-resolved total and projected densities of states (TDOS and PDOS) as shown in Figure 5. The TDOS displays a pronounced asymmetry between the spin-up and spin-down channels, evidencing strong intrinsic spin polarization. This behavior originates from the Fe-centered octahedral environment, where the Br-ligand crystal field splits the Fe 3$d$ manifold and, in the presence of magnetic exchange interactions, produces sizeable spin-dependent energy shifts of Fe-dominated bands[69,70].

The PDOS further clarifies the microscopic origin of the band edges. In the valence region close to the Fermi level, Br 4$p$ states provide the dominant contribution, with Fe 3$d$ states mixing to a spin-dependent degree through Fe–Br covalency. The stronger Fe 3$d$–Br 4$p$ hybridization lowers the spin-up valence states relative to the spin-down channel, consistent with an exchange-split $d$ manifold under crystal-field control. At the conduction-band onset, Fe-derived states become increasingly important, while the alkali states (Na and Cs) contribute mainly at higher energies, reflecting their more ionic, weakly hybridized character. Overall, the DOS indicates that the fundamental excitation involves transitions from Br 4$p$-dominated valence states into Fe-centered conduction states, implying that band-edge physics is governed primarily by the $FeBr_6$ octahedral framework. The spin-selective band-edge alignment inferred from the DOS is therefore a direct consequence of (i) exchange splitting within the Fe 3$d$ states and (ii) asymmetric Fe–Br $p$–$d$ hybridization between the two spin channels. This combination naturally leads to different effective band-edge energies for the two spin manifolds and supports the generation of spin-polarized carriers under electrical bias or optical excitation, which is central to spintronic and spin-optoelectronic concepts[67,68].

Including on-site correlation via the Hubbard $U$ correction on Fe 3$d$ states further sharpens this picture. The $U$ term reduces the self-interaction error and enhances the localization of Fe $d$ electrons, pushing occupied Fe-$d$ features to lower energies and unoccupied $d$ features upward, thereby suppressing spurious near-edge Fe–Br overlap typical of semilocal functionals. As a result, the spin polarization around the band edges is preserved and typically enhanced, yielding a cleaner description of spin-polarized semiconductivity consistent with correlated transition-metal halide perovskites[69,70]. In summary, the DOS analysis demonstrates that the electronic structure of $Cs_2NaFeBr_6$ is governed by Br 4$p$ valence states and exchange-split Fe-derived conduction states, with spin-dependent $p$–$d$ hybridization underpinning its potential for spin-selective transport and spin-photovoltaic functionality[66–68].

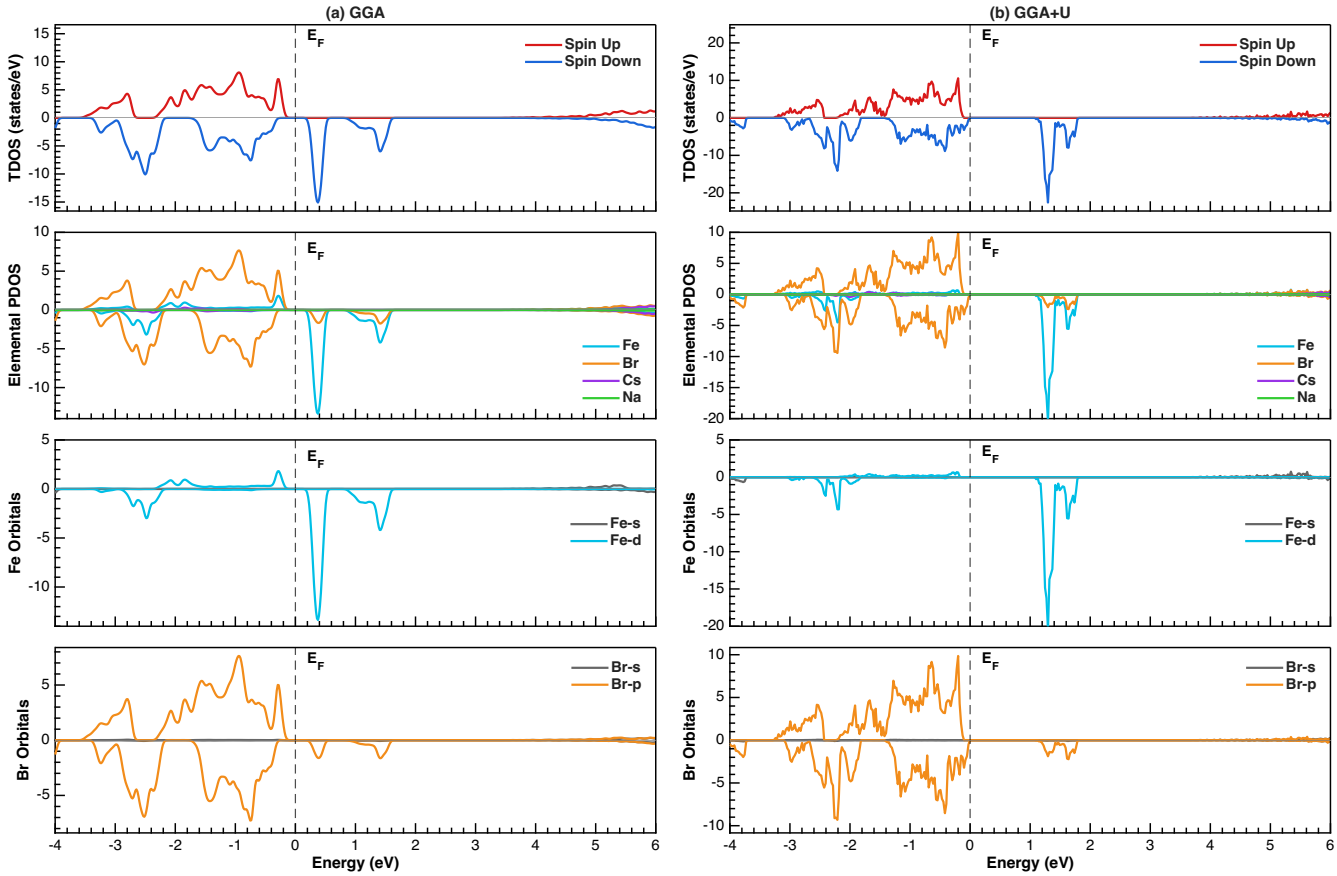


**Figure 5.** Total and atom-/orbital-resolved density of states (DOS) of $Cs_2NaFeBr_6$ calculated using (a) GGA and (b) GGA+$U$. The Fermi level is set to 0 eV.

### *Carrier Effective Mass*

Carrier effective masses provide a direct measure of the band-edge curvature and are therefore central to assessing the photovoltaic and optoelectronic potential of $Cs_2NaFeBr_6$. In particular, the electron and hole effective masses strongly influence carrier mobility, electrical resistivity, and the free-carrier contribution to the optical response. In this work, the effective masses were estimated by fitting a second-order (parabolic) function to the band dispersion in the vicinity of the conduction-band minimum (CBM) and valence-band maximum (VBM), using

$$m^* = \hbar^2 \left[ \frac{\partial^2 E(k)}{\partial k^2} \right]^{-1}, \tag{8}$$

where $k$ is the wave-vector component along a given high-symmetry direction, $\hbar$ is the reduced Planck constant, and $E(k)$ denotes the corresponding band energy.

The extracted band-edge curvatures indicate that the electron effective mass at the CBM is relatively small, with $m_e^*$ lying in the range $\sim$0.766–0.820 $m_0$ along the considered symmetry directions. Such a moderate-to-low electron mass implies comparatively high electron mobility and favorable charge-transport capability, which is beneficial for photovoltaic and photocatalytic operation where rapid extraction of photogenerated electrons is desirable. In contrast, the hole effective mass at the VBM is larger, with $m_h^*$ spanning $\sim$0.801–0.981 $m_0$, consistent with a flatter valence-band dispersion and weaker band-edge curvature. This electron–hole mass asymmetry suggests that electron transport should be more efficient than hole transport in $Cs_2NaFeBr_6$.

Overall, the coexistence of a comparatively low electron effective mass, a larger hole effective mass, and the spin-polarized electronic structure discussed in the preceding section points to an indirect-gap semiconductor with asymmetric carrier-transport characteristics. From an application perspective, the lighter electron mass supports efficient electron conduction, while the heavier hole mass may reduce the overlap of electron and hole transport channels and can contribute to suppressing carrier recombination, thereby aiding the separation and collection of photogenerated carriers in photovoltaic-type devices.

## Optical and Dielectric Properties

Motivated by the electronic-structure findings, a detailed investigation of the optical response of the double perovskite $Cs_2NaFeBr_6$ was performed to elucidate its light–matter interaction and optoelectronic potential. Optical properties are essential for understanding how a material absorbs, reflects, and transmits electromagnetic radiation, thereby indicating its suitability for device integration. In this work, the principal optical functions of $Cs_2NaFeBr_6$ including the complex dielectric function, energy loss spectrum, refractive index, reflectivity, optical conductivity, and absorption coefficient were calculated in the photon-energy range of 0–25 eV.

The complex dielectric function, $\varepsilon(\omega)$, is the most fundamental optical quantity from which other optical constants can be derived[50]. It is expressed as[45]

$$\varepsilon(\omega) = \varepsilon_1(\omega) + i\varepsilon_2(\omega),$$

where the real part $\varepsilon_1(\omega)$ is associated with polarization (stored energy) and the imaginary part $\varepsilon_2(\omega)$ represents energy dissipation arising from electronic excitations[71]. The calculated dielectric spectra of $Cs_2NaFeBr_6$ are shown in Fig. 6(a). The static dielectric constant is obtained as $\varepsilon_1(0) \approx 6.5$, indicating strong polarizability at low photon energies. The dependence of the static dielectric constant on the optical band gap $E_g$ can be described by Penn's model[72]:

$$\varepsilon_1(0) = 1 + \left(\frac{\hbar\omega_p}{E_g}\right)^2.$$

In semiconductors, interband transitions make the dominant contribution to $\varepsilon(\omega)$, whereas intraband effects are more relevant for metallic systems[73]. For $Cs_2NaFeBr_6$, $\varepsilon_2(\omega)$ begins to rise from $\sim$1.5 eV, implying that this energy is sufficient to initiate electronic transitions from the valence band to the conduction band and supporting a narrow-gap semiconducting character[74]. In the intermediate photon-energy region, $\varepsilon_1(\omega)$ exhibits multiple oscillations followed by a gradual reduction, reflecting weaker polarization at higher energies. Meanwhile, $\varepsilon_2(\omega)$ shows prominent peaks (notably in the 5–7 eV region), which correspond to strong interband optical transitions. Importantly, $\varepsilon_1(\omega)$ approaches and crosses zero around 12–14 eV, indicating the bulk plasma frequency. At energies above this region, $\varepsilon_1(\omega)$ becomes negative, implying a metallic-like optical response and enhanced reflection at high photon energies[75], while $\varepsilon_2(\omega)$ gradually decays, indicating reduced absorption.

The static dielectric response further supports the strong polarizability of $Cs_2NaFeBr_6$. The calculated static dielectric constant is $\varepsilon_1(0) \approx 6.50$, which is substantially higher than those of the related $Cs_2NaTiX_6$ compounds ($\varepsilon_1(0) = 3.13$ for X = Br, 2.48 for X = Cl, and 1.93 for X = F). This comparatively larger $\varepsilon_1(0)$ can be attributed to the enhanced polarizability associated with heavier halogen ions, which strengthens dielectric screening and can suppress charge-carrier recombination which is an advantageous feature for optoelectronic device performance[76]. In line with this behavior, $Cs_2NaFeBr_6$ is expected to show relatively lower transmittance in the low-energy region where absorption is pronounced, while higher transmittance is anticipated at higher photon energies as the absorption gradually decreases.

The energy loss function describes the energy lost by fast electrons interacting with the electronic structure of the material (primarily through plasmon excitations and interband transitions)[77]. It is given by

$$L(\omega) = \frac{\varepsilon_2(\omega)}{\varepsilon_1^2(\omega) + \varepsilon_2^2(\omega)}.$$

As shown in Fig. 6(b), $Cs_2NaFeBr_6$ exhibits a dominant loss peak at $\sim$22.5 eV, indicating strong collective oscillations of electrons (plasmon response). In addition to plasmon excitations, peaks in $L(\omega)$ may also be influenced by other mechanisms such as band-to-band transitions, phonon excitations, and inner-shell ionization[78].

To describe light propagation through crystalline materials, an understanding of optical refraction is essential, as the trajectory of light bends closer to the surface normal with increasing refractive index[50]. The refractive index $n(\omega)$ is therefore a key optical parameter, as it governs how incident photons interact with the lattice and also provides insight into the nature of chemical bonding within the material[79]. In general, a higher refractive index is associated with stronger covalent bonding, whereas lower values indicate more ionic character[80].

The optical response is described by the complex refractive index, defined as[81]

$$n'(\omega) = n(\omega) + ik(\omega),$$

where $n(\omega)$ is the refractive index and $k(\omega)$ is the extinction coefficient, which accounts for optical attenuation. These quantities can be derived directly from the complex dielectric function according to

$$n(\omega) = \frac{1}{\sqrt{2}}\left[\sqrt{\varepsilon_1^2(\omega) + \varepsilon_2^2(\omega)} + \varepsilon_1(\omega)\right]^{1/2}, \qquad k(\omega) = \frac{1}{\sqrt{2}}\left[\sqrt{\varepsilon_1^2(\omega) + \varepsilon_2^2(\omega)} - \varepsilon_1(\omega)\right]^{1/2}.$$

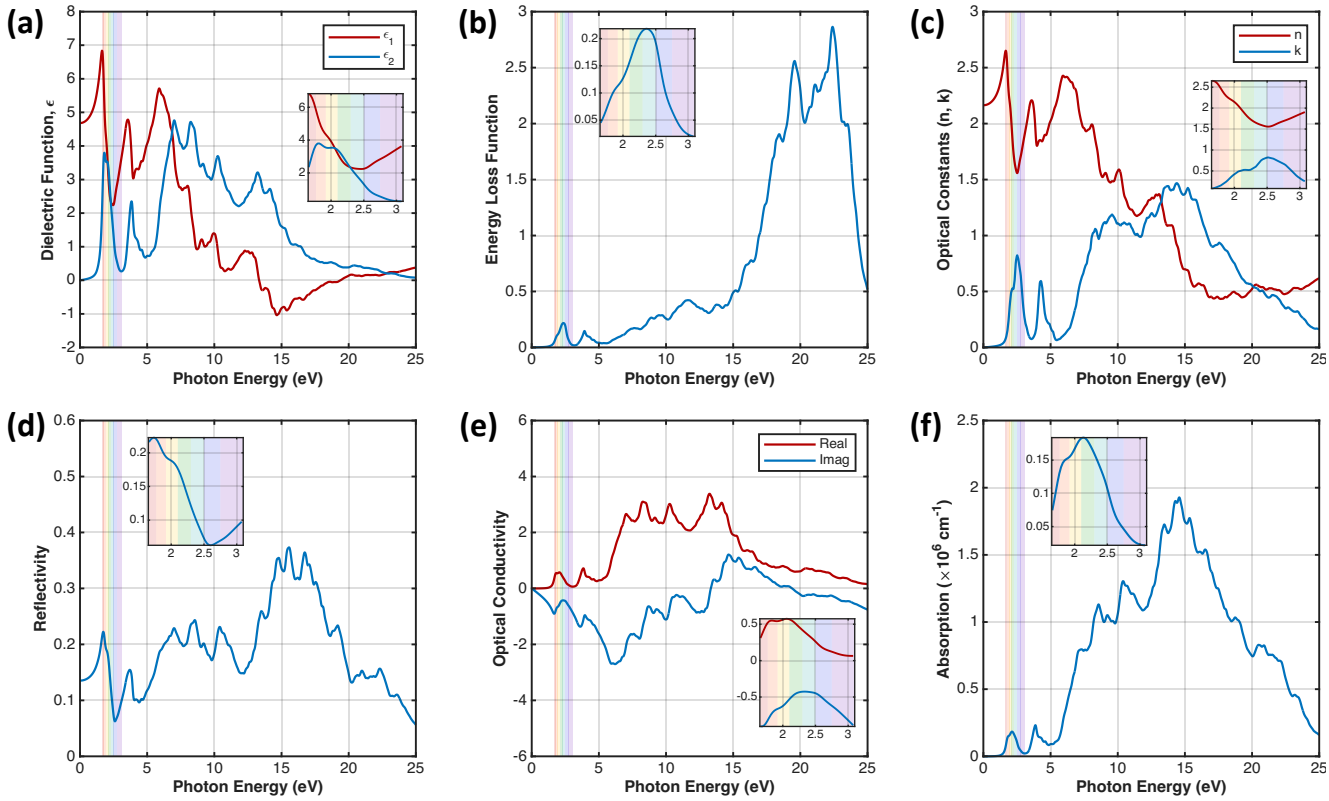


**Figure 6.** Calculated optical properties of $Cs_2NaFeBr_6$: (a) real and imaginary parts of the dielectric function, (b) energy loss spectrum, (c) refractive index, (d) reflectivity, (e) optical conductivity, and (f) absorption coefficient as a function of photon energy.

As shown in Fig. 6(c), the static refractive index of $Cs_2NaFeBr_6$ is $n(0) \approx 2.62$, which satisfies the relation $n^2(0) = \varepsilon_1(0)$ and confirms the internal consistency of the optical calculations. The extinction coefficient closely follows the behavior of the imaginary part of the dielectric function $\varepsilon_2(\omega)$, indicating that optical attenuation is primarily governed by interband electronic transitions responsible for dielectric loss. The relatively high refractive index of $Cs_2NaFeBr_6$ suggests a notable degree of covalent character in its bonding nature, which is favorable for optoelectronic and photonic applications[80].

Reflectivity spectra are crucial for optoelectronic device design because they describe how light behaves at a material interface and, consequently, influence light penetration, absorption, and transmission, as well as the engineering of coatings and reflective layers. Reflectivity represents the fraction of incident radiation reflected from the surface and is governed by factors such as refractive index, electronic band structure, and surface condition. It can be obtained from the complex refractive index $n'(\omega)$ as[82]

$$R(\omega) = \left| \frac{n'(\omega) - 1}{n'(\omega) + 1} \right|^2 .$$

In general, high reflectivity is often associated with metallic-like optical behavior, whereas lower reflectivity is typical of semiconductors and insulators[83]. As shown in Fig. 6d, $Cs_2NaFeBr_6$ exhibits its maximum reflectivity in the 12.5–18 eV range, indicating a strong tendency to reflect radiation in this ultraviolet energy window. The reflectivity initially increases from 0 to ∼2 eV, suggesting enhanced surface reflection as the photon energy rises, which may be related to the onset of optical transitions within the material[84]. After this initial rise, the reflectivity decreases and reaches a minimum near ∼2.5 eV, consistent with stronger photon absorption that reduces the reflected intensity. At higher energies, the spectrum becomes oscillatory, reflecting dispersion effects and multiple interband transitions, and reaches a pronounced maximum around ∼16.25 eV. Overall, these features indicate that $Cs_2NaFeBr_6$ retains semiconducting character at low photon energies but develops a metallic-like optical response at higher energies, implying stronger potential for ultraviolet optoelectronic applications rather than efficient light harvesting in the visible region.

The optical conductivity $\sigma(\omega)$ arises from intraband and interband electronic processes[85]. As shown in Fig. 6(e), $\sigma(\omega)$ increases from the origin and exhibits strong activity in the 5–10 eV region, reaching a maximum near $\sim$14 eV before decreasing with further oscillations. These features reflect the underlying electronic structure and available optical transition channels; in particular, contributions from Cs-related states can significantly influence the optical response in this energy range.

The absorption coefficient quantifies photon attenuation inside the material and is expressed as[83]

$$\alpha(\omega) = \frac{2\omega}{c} k(\omega).$$

The absorption spectrum of $Cs_2NaFeBr_6$ (Fig. 6(f)) shows that notable absorption begins around $\sim$4 eV, indicating the onset of significant electronic transitions. Several absorption peaks appear across the spectrum, and the maximum absorption coefficient reaches nearly $2.0 \times 10^6$ (in the plotted units) around $\sim$15 eV. As typically observed in semiconductors, absorption decreases in the higher-energy region after the strong-transition regime. Overall, the computed optical response, particularly the strong ultraviolet absorption, pronounced dielectric screening, and tunable reflectivity highlights $Cs_2NaFeBr_6$ as a promising candidate for UV-active optoelectronic and photonic applications, with potential relevance to energy-harvesting technologies.

### Thermoelectric Coefficients

In this study, the thermoelectric transport coefficients were computed within the linearized semiclassical Boltzmann transport equation as implemented in BoltzTraP2[86]. For spin-polarized calculations, BoltzTraP2 does not perform separate thermoelectric transport calculations for spin-up and spin-down channels; instead, it evaluates the transport coefficients from the summed contribution of both spin channels. Within the relaxation-time approximation, BoltzTraP2 defines the transport distribution function

$$\sigma(\varepsilon, T) = \int \sum_b \mathbf{v}_{b,\mathbf{k}} \otimes \mathbf{v}_{b,\mathbf{k}} \, \tau_{b,\mathbf{k}} \, \delta\left(\varepsilon - \varepsilon_{b,\mathbf{k}}\right) \frac{d\mathbf{k}}{8\pi^3},$$

and evaluates the moments (generalized transport coefficients)

$$L^{(\alpha)}(\mu; T) = q^2 \int \sigma(\varepsilon, T) \, (\varepsilon - \mu)^\alpha \left( -\frac{\partial f^{(0)}(\varepsilon; \mu, T)}{\partial \varepsilon} \right) d\varepsilon,$$

where $f^{(0)}$ is the Fermi–Dirac distribution, $\mu$ is the chemical potential, $q$ is the carrier charge, and $\tau_{b,\mathbf{k}}$ is the relaxation time (taken as a constant $\tau_0$ under CRTA). The electrical conductivity, Seebeck coefficient, and electronic thermal conductivity (charge-carrier contribution) are then obtained as

$$\sigma = L^{(0)}$$

$$S = \frac{1}{qT} \frac{L^{(1)}}{L^{(0)}}$$

$$\kappa_e = \frac{1}{q^2 T} \left[ \frac{\left(L^{(1)}\right)^2}{L^{(0)}} - L^{(2)} \right]$$

The thermoelectric transport coefficients were evaluated as a function of chemical potential relative to the Fermi level $(\mu - E_F)$ at 300 K, 500 K and 700 K within the semiclassical Boltzmann transport formalism under the constant relaxation-time approximation. In this representation, negative chemical potential corresponds to p-type (hole-dominated) conditions, whereas positive chemical potential represents n-type (electron-dominated) conditions. The calculated Seebeck coefficient (S) exhibits pronounced extrema in the vicinity of the band edges, characterized by a strong positive thermopower peak on the p-type side just below $\mu - E_F = 0$ and a deep negative minimum immediately on the n-type side, followed by a gradual reduction in $|S|$ as $|\mu - E_F|$ increases. Quantitatively, the maximum p-type thermopower reaches $+1550.31\ \mu\mathrm{V\,K^{-1}}$ at 300 K, decreasing to $+1056.36\ \mu\mathrm{V\,K^{-1}}$ at 500 K, and further to $+770.57\ \mu\mathrm{V\,K^{-1}}$ at 700 K. On the n-type side, the corresponding minima are $-1550.31\ \mu\mathrm{V\,K^{-1}}$ at 300 K, $-958.06\ \mu\mathrm{V\,K^{-1}}$ at 500 K, and $-657.21\ \mu\mathrm{V\,K^{-1}}$ at 700 K. Notably, the absolute thermopower $|S|$ decreases systematically with increasing temperature. This behaviour can be rationalized by considering how temperature reshapes the population of carriers that participate in transport: at low temperature, only a narrow set of electronic states very close to the chemical potential contributes to charge flow, so transport is dominated by carriers within a tight energy window near the band edge. In this regime, the conducting carriers are effectively "energy-selected", and the current is carried

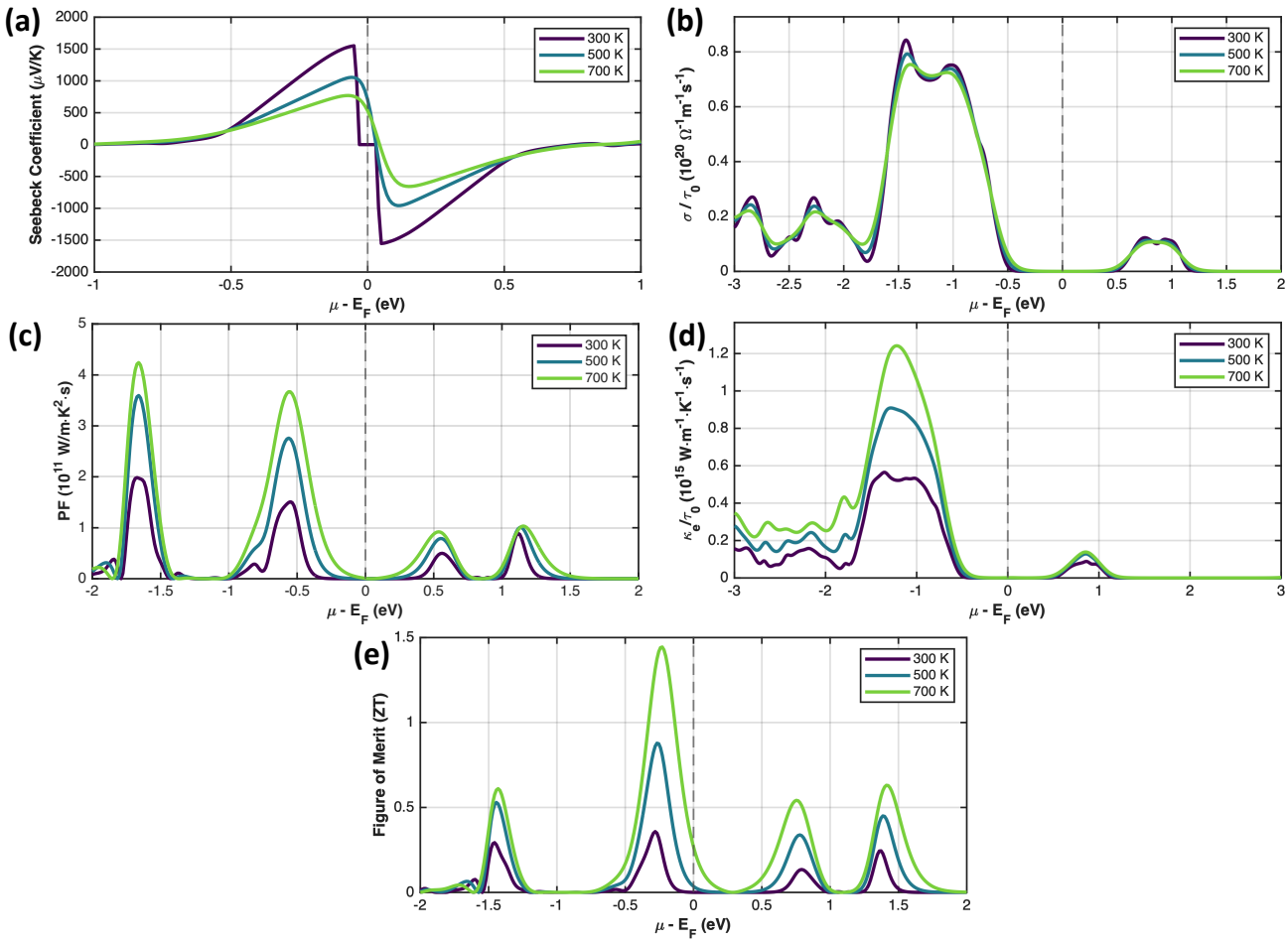


**Figure 7.** Chemical-potential dependence of the thermoelectric transport coefficients at 300 K, 500 K, and 700 K: (a) Seebeck coefficient $S$ ($\mu$V K$^{-1}$), (b) electrical conductivity $\sigma/\tau_0$ ($10^{20}$ $\Omega^{-1}$ m$^{-1}$ s$^{-1}$), (c) electronic thermal conductivity $\kappa_e/\tau_0$ ($10^{15}$ W m$^{-1}$ K$^{-1}$ s$^{-1}$), and (d) dimensionless figure of merit $ZT$, plotted as functions of $\mu - E_F$ (eV). The vertical dashed line marks $\mu = E_F$.

predominantly by carriers whose energies lie preferentially on one side of $\mu$, producing a large imbalance between the average transport energy and $\mu$ and hence a large thermopower. As temperature rises, the Fermi–Dirac distribution broadens, so states over a wider energy range around $\mu$ become partially occupied (or partially emptied) and begin contributing to conduction. This thermal broadening increases the number of active carriers and expands the contributing energy window from a sharp band-edge region to a broader portion of the band structure. Consequently, transport becomes less selective: carriers with energies both closer to and farther from $\mu$ contribute simultaneously, reducing the net asymmetry in energy-dependent transport and lowering the average transported energy relative to $\mu$. Because the Seebeck coefficient fundamentally measures this energy asymmetry (i.e., the entropy carried per unit charge), the reduction in energy-selective transport at elevated temperatures leads directly to a decrease in $|S|$, even though electrical conductivity may increase due to the larger carrier population.

The electrical conductivity per relaxation time, $\sigma/\tau_0$, is strongly suppressed around $\mu - E_F \approx 0$, consistent with the chemical potential lying within the gap where the density of available conducting states is minimal. As $\mu$ is displaced into the valence or conduction bands, $\sigma/\tau_0$ increases sharply and exhibits multiple peaks reflecting the detailed band dispersion and associated carrier velocities. Across the investigated temperature range, the overall profile remains qualitatively similar, indicating that conductivity is primarily governed by the availability of electronic states and the carrier population set by $(\mu - E_F)$, while temperature mainly broadens the energy window of contributing carriers. To assess the combined influence of thermopower and conductivity, the power factor $PF = S^2\,\sigma/\tau_0$ was computed. The resulting PF curves display well-defined maxima most prominently in the p-type region, demonstrating that optimal performance arises from a balance between large $|S|$ and sufficiently high $\sigma/\tau_0$. Secondary maxima are also observed under electron-doped (n-type) conditions, indicating additional operating windows depending on the achievable carrier tuning.

The electronic thermal conductivity per relaxation time, $\kappa_e/\tau_0$, broadly follows the trend of $\sigma/\tau_0$ because both are governed by the same transport distribution (state availability and carrier velocities), differing mainly in their weighting of higher-energy carriers. Consequently, $\kappa_e/\tau_0$ is strongly suppressed near $\mu - E_F \approx 0$ (band gap region with minimal mobile carriers) and increases rapidly as $\mu$ shifts into the valence or conduction bands. Moreover, $\kappa_e/\tau_0$ increases markedly with temperature due to thermal broadening, which activates a wider range of carrier energies around $\mu$ and enhances electronic heat transport. Within the plotted window, the maximum $\kappa_e/\tau_0$ occurs in the p-type region and rises from $0.564771 \times 10^{15}$ $\mathrm{W\,m^{-1}\,K^{-1}\,s^{-1}}$ at 300 K, to $0.909093 \times 10^{15}$ $\mathrm{W\,m^{-1}\,K^{-1}\,s^{-1}}$ at 500 K, and to $1.24156 \times 10^{15}$ $\mathrm{W\,m^{-1}\,K^{-1}\,s^{-1}}$ at 700 K. Notably, these maxima occur deeper in the valence band than the Seebeck extrema, reflecting the expected trade-off: $|S|$ peaks near band edges where energy-selective transport is strongest, whereas $\kappa_e/\tau_0$ is maximized under heavier doping where carrier population and band velocities are large. Thus, the chemical-potential regions maximizing $\kappa_e/\tau_0$ generally do not coincide with those optimizing thermopower, implying that the best thermoelectric performance is achieved in an intermediate doping window where $|S|$ remains high while electronic heat leakage is limited.

Finally, the dimensionless thermoelectric figure of merit, $zT$, was analyzed as a function of chemical potential at representative temperatures. In this study, $zT$ is calculated using the following expression:

$$zT = \frac{S^2 \sigma T}{\kappa_e + \kappa_l} \tag{12}$$

It is important to emphasize that while the BoltzTraP2 code provides an accurate first-principles framework for calculating electronic transport coefficients ($S$, $\sigma$, and $\kappa_e$)[86], it focuses solely on the electronic contributions and does not calculate the lattice thermal conductivity, $\kappa_l$, which arises from phonon-phonon scattering. Consequently, a constant value of $\kappa_l = 1.0$ $\mathrm{W\,m^{-1}\,K^{-1}}$ was adopted in our $zT$ evaluation. This assumption is supported by extensive literature on lead-free halide double perovskites, which consistently identifies these materials as possessing ultralow lattice thermal conductivity. Specifically, all-inorganic halide perovskites exhibit strong phonon resonance scattering and dynamic cation off-centering, leading to thermal conductivities as low as 0.32 $\mathrm{W\,m^{-1}\,K^{-1}}$[87]. Research on various Cs-based double perovskites, such as $Cs_2MScCl_6$ (M = Na, K)[88] and $Cs_2KTlX_6$[89], confirms that these structures naturally possess low thermal conductance and minimal thermal conductivity. Furthermore, lead-free double perovskites like $Cs_2BiAgBr_6$ and $Cs_2BiAgCl_6$ have demonstrated extremely low $\kappa_l$ values ranging from 0.103 to 0.352 $\mathrm{W\,m^{-1}\,K^{-1}}$[90]. Studies on $Cs_2NaBiCl_6$[22], $Cs_2AuAsX_6$[91], and $Cs_2CuBiX_6$[92] similarly highlight that the heavy-atom rattling and structural complexity of these halide systems are conducive to high thermoelectric performance via suppressed heat transport. Therefore, utilizing $\kappa_l = 1.0$ $\mathrm{W\,m^{-1}\,K^{-1}}$ serves as a conservative upper-bound estimate for $zT$.

The calculated $zT(\mu)$ curves exhibit sharp enhancements at chemical potentials where the power factor is maximized, and the peak heights increase markedly with temperature, consistent with the $zT \propto S^2\sigma T/\kappa$ dependence. A dominant maximum is observed near chemical potentials close to the intrinsic reference, while additional local maxima appear on both the p-type and n-type sides, reflecting the presence of multiple favourable transport windows tied to band-edge features. Overall, these results highlight that high thermoelectric performance is achieved only within narrow chemical-potential ranges near the band edges, emphasizing the importance of targeted carrier tuning to place $\mu$ in the regime where the electronic structure simultaneously supports large thermopower and strong electrical transport.

## Power Conversion Efficiency

Photovoltaic (PV) energy conversion is among the most sustainable approaches for meeting the growing global energy demand. For single-junction absorbers under AM 1.5G illumination, the optimal bandgap typically lies in the $\sim$ 0.8–1.8 eV range, where the Shockley–Queisser (S–Q) detailed-balance limit can exceed 25%[93]. Beyond bandgap alignment, high optical absorption across the visible–near-infrared region, high carrier mobility, and low exciton binding energy are essential to maximize device performance. In particular, a large absorption coefficient ($> 10^4$ $\mathrm{cm^{-1}}$) enables efficient photon harvesting in sub-micrometre films, thereby reducing carrier transport distances and mitigating bulk recombination. Low exciton binding energy facilitates spontaneous electron–hole separation, while high mobility supports rapid charge extraction prior to recombination.

The proposed halide double perovskite $Cs_2NaFeBr_6$ exhibits a direct bandgap of 1.09 eV, close to crystalline silicon ($\sim$ 1.1 eV), placing it near the optimum of the S–Q efficiency curve. For this bandgap, the S–Q model predicts a maximum efficiency of $\sim$ 32.2% at 298.15 K under AM 1.5G illumination. However, the S–Q limit depends primarily on the bandgap and does not explicitly account for material-specific optical absorption and transition characteristics, which can lead to optimistic estimates.

To obtain a more realistic upper bound, we employed the spectroscopic limited maximum efficiency (SLME) formalism, which incorporates the calculated absorption spectrum and the nature of the electronic transition. The thickness-dependent short-circuit current density ($J_{sc}$), open-circuit voltage ($V_{oc}$), and power conversion efficiency (PCE) are summarized in Figure 8 and Table 4. The calculated $J_{sc}$ increases rapidly with absorber thickness and approaches saturation beyond $\sim 1$ $\mu$m, consistent

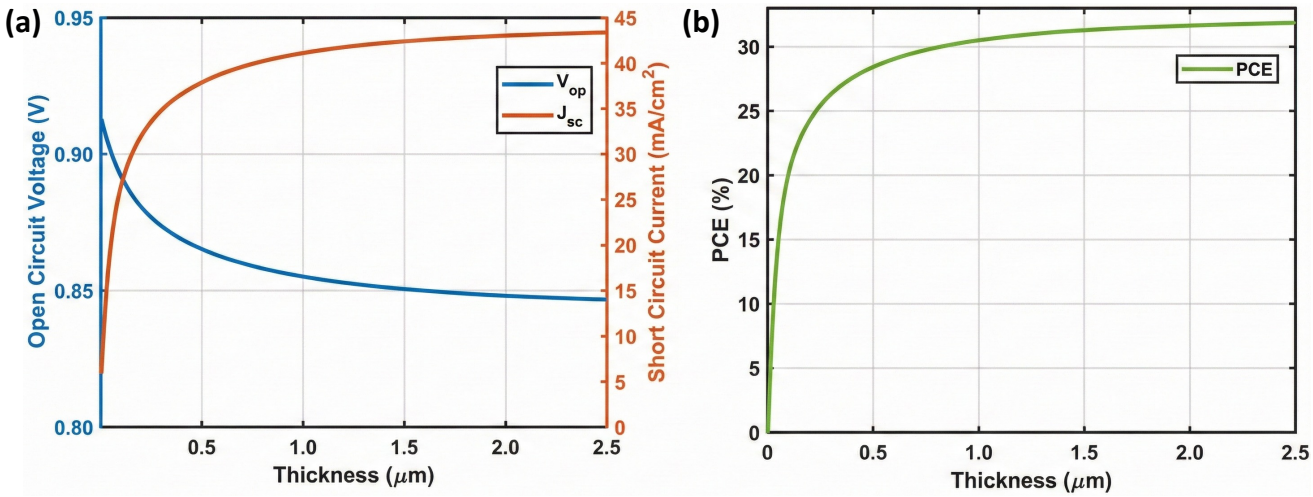


**Figure 8.** (a) Short-circuit current density ($J_{sc}$) and open-circuit voltage ($V_{oc}$) as a function of absorber-layer thickness. (b) Spectroscopic limited maximum efficiency (SLME) as a function of absorber-layer thickness.

with enhanced photon harvesting. At 1 $\mu$m, $J_{sc}$ reaches 41.26 mA cm$^{-2}$ and further increases to 43.88 mA cm$^{-2}$ at 5 $\mu$m, approaching the S–Q value ($\sim$ 44.2 mA cm$^{-2}$). In contrast, $V_{oc}$ decreases slightly with thickness (from 0.855 V at 1 $\mu$m to 0.844 V at 5 $\mu$m), consistent with increased bulk recombination in thicker films. Consequently, the SLME-derived PCE is 30.60% at 1 $\mu$m and reaches a maximum of 32.09% at 5 $\mu$m, approaching the S–Q limit.

**Table 4.** (A) Shockley–Queisser (S–Q) limit and spectroscopic limited maximum efficiency (SLME) metrics for $Cs_2NaFeBr_6$ under AM 1.5G illumination. (B) Benchmark comparison of experimental and theoretical (SLME) power conversion efficiencies for selected perovskite absorbers.

| **(A) $Cs_2NaFeBr_6$: S–Q limit vs. SLME** | | | |
|---|---|---|---|
| **Parameter** | **S–Q**[93] | **SLME simulated** | |
| | | **at 1 $\mu$m** | **at 5 $\mu$m (max PCE)** |
| $J_{sc}$ (mA cm$^{-2}$) | $\sim$ 44.2 | 41.26 | 43.88 |
| $V_{oc}$ (V) | $\sim$ 0.84 | 0.855 | 0.844 |
| PCE (%) | $\sim$ 32.2 | 30.60 | 32.09 |
| **(B) Benchmark: Experimental vs. SLME theoretical PCE** | | | |
| **Perovskite** | **Experimental PCE (%)** | | **Theoretical PCE (%) (SLME)** |
| $FAPbI_3$ | 26.52[94], 26.25[95] | | 31.82[96] |
| $MAPbI_3$ | 21.25[97], 20.24[98] | | 29.00[99] |
| $CsPbI_3$ | 22.05[100], 21.08[101] | | 30.41[102] |
| $Cs_2AgBiBr_6$ | 4.23[103], 6.37[104] | | 7.92[105] |

## Computational Methodology

First-principles calculations were carried out within the framework of Density Functional Theory (DFT) using the Vienna *Ab Initio* Simulation Package (VASP). The electron–ion interaction was described by the projector-augmented wave (PAW) method as implemented in VASP. The valence electron configurations employed for the constituent elements were Cs ($5s^2 5p^6 6s^1$), Na ($2s^2 2p^6 3s^1$), Fe ($3d^6 4s^2$), and Br ($4s^2 4p^5$). Exchange–correlation effects were treated using the generalized gradient approximation (GGA) with the revised Perdew–Burke–Ernzerhof functional for solids (PBEsol), which is suitable for structural properties of crystalline materials. To properly capture the localized nature of the Fe-3d states and the magnetic character of $Cs_2NaFeBr_6$, the GGA+$U$ approach was adopted. An effective on-site Coulomb interaction of $U_{eff} = 3$ eV was applied to the

Fe-3d orbitals, following prior reports that demonstrated reliable agreement for Fe-based halide systems[106]. Spin-polarized calculations were performed by initializing the Fe atoms with an initial magnetic moment of 5 $\mu_B$ per atom, while all other atoms were initialized with 0 $\mu_B$. A plane-wave kinetic energy cutoff of 600 eV was used for all computations. Brillouin-zone integrations were conducted using a Monkhorst–Pack $k$-point grid of $6 \times 6 \times 6$ for structural relaxation, and a denser $9 \times 9 \times 9$ mesh for electronic structure, magnetic, elastic, and related property calculations. Structural optimization was continued until the residual Hellmann–Feynman force on each atom was below 0.005 eV/Å and the total-energy difference between consecutive ionic steps was less than $10^{-8}$ eV. Phonon properties were evaluated using density functional perturbation theory (DFPT) as implemented in VASP, and the force-constant post-processing and phonon dispersion calculations were performed using the PHONOPY package.

## Conclusion

In summary, we employed first-principles calculations to evaluate the lead-free halide double perovskite $Cs_2NaFeBr_6$ with a particular focus on stability, fundamental electronic behavior, optical response, and photovoltaic promise. The structural formability indicators and thermodynamic considerations collectively suggest that the cubic phase is favorable, while phonon analysis further verifies dynamical stability, supporting the feasibility of this compound beyond purely geometric arguments. The calculated elastic response also confirms mechanical stability and points to a comparatively compliant halide lattice, which is often desirable for thin-film processing and can be beneficial for mitigating strain accumulation and defect-induced performance losses in practical devices. From an electronic standpoint, $Cs_2NaFeBr_6$ exhibits a direct-gap semiconducting character with band-edge features that can support effective photoexcitation and carrier transport, and the associated dielectric response indicates meaningful screening, which is relevant for suppressing Coulombic trapping and nonradiative recombination pathways. In addition, the optical properties demonstrate strong interaction with light in the technologically relevant spectral region, reinforcing its suitability as an absorber layer. Most importantly, the SLME assessment predicts a high theoretical photovoltaic efficiency that approaches the Shockley–Queisser ceiling for the calculated bandgap, highlighting the potential of $Cs_2NaFeBr_6$ as a competitive lead-free alternative for next-generation solar absorbers. Overall, the combined evidence from structural, dynamical, mechanical, electronic, and optical analyses provides a consistent picture of a stable and optoelectronically active double perovskite, and it motivates future experimental efforts to synthesize high-quality films, characterize defect physics and carrier lifetimes, and validate device performance under realistic operating conditions.

## Data availability

The data sets generated and thereafter analysed would be available from the corresponding author upon reasonable request.

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## Acknowledgements

The authors are grateful to the Bangladesh University of Engineering and Technology (BUET) for providing the necessary research facilities and to the Committee for Advanced Studies and Research (CASR), BUET, for partial financial support. Additionally, we sincerely thank Dr. Shaestagir Chowdhury for providing computational resources.

## Author contributions statement

M.A.R. and M.J.I. contributed equally as co-first authors and performed the calculations and post-processing. F.G. conceived and supervised the study. M.A.R., M.J.I., T.C., M.R.U., and A.R. wrote the initial drafts and analyzed the results. All authors reviewed and approved the manuscript.

## Competing Interests

The authors declare no competing interests.

## Additional information

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